

UNIT 3 ASSESSMENT OF LEARNING: CURVE SKETCHING – DAY 1

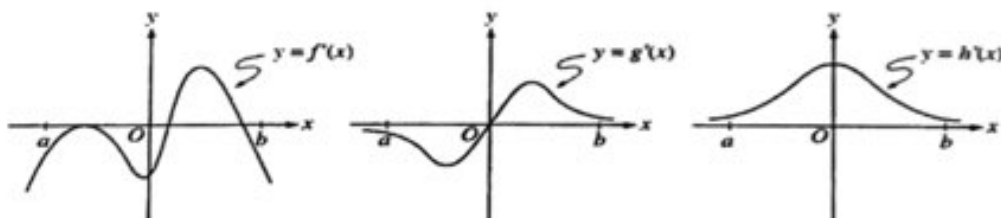
Instructions: You MUST use concepts covered in this unit/course. Leave all answers EXACT unless otherwise indicated. Show all steps for full marks. The use of cellphones, audio or video recording devices, digital music players or email or text-messaging devices during the assessment is prohibited.

K & U	Thinking	Comm.
/16	/5	/6

KNOWLEDGE & UNDERSTANDING - [16 Marks]

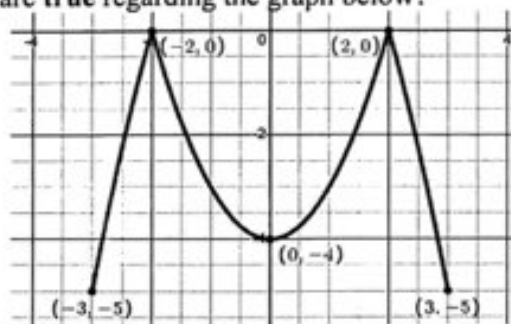
Multiple Choice: Write the CAPITAL LETTER corresponding to the correct answer on the line provided.
[6 Marks]

- The function $f(x) = -\frac{1}{3}x^3 + x - 31$ is decreasing on the interval(s) C
 A. $(-\infty, -1)$ B. $(-\infty, \infty)$ C. $(-\infty, -1) \cup (1, \infty)$ D. $(-1, 1)$
- Which of the following is/are true? B
 I. The function $f(x) = \frac{x-5}{x^3-8}$ does not have a cusp.
 II. If $f'(c) = 0$ and $f''(c) > 0$ then $f(x)$ has a local minimum $x = c$.
 III. The function $f(x) = \frac{x^2-4}{x^2-1}$ has critical points at $x = \pm 1$.
 IV. If a function has a vertical tangent at $x = -\frac{1}{7}$, then there is a point of inflection at $x = -\frac{1}{7}$.
 A. I and III only B. I, II and IV only C. II, III and IV only D. All are true.
- The critical number(s) of $f(x) = 4x^5 + 5x^3$ are C
 A. $0, \pm \frac{\sqrt{3}}{2}$ B. $\pm \frac{\sqrt{3}}{2}$ C. 0 D. There are none.
- The graph of the derivatives of $f(x)$, $g(x)$, and $h(x)$ are shown below. A



Which of the function(s) have a local maximum on $x \in (a, b)$?

- Which of the following is/are true regarding the graph below? C



- There is an absolute minimum at $(0, -4)$ and local maximums at $(-2, 0)$ and $(2, 0)$.
 II. There is a local minimum at $(0, -4)$ and absolute minimums at $(-3, -5)$ and $(3, -5)$.
 III. There are local minimums at $(0, -4)$, $(-3, -5)$ and $(3, -5)$.
 IV. Points $(-2, 0)$ and $(2, 0)$ are both local and absolute maximums.
 A. I and IV only B. II and III only C. II and IV only D. I, II and IV only
- The value of the constant b for the function $f(x) = x^2 + bx + 4$, which has a critical point at $x = -2$ is C
 A. 1 B. 2 C. 4 D. 5

7. For the function $f(x) = -\frac{9}{2}x^{\frac{2}{3}} + 9x^{\frac{1}{3}}$, determine the **exact** coordinates of the point(s) of inflection **and** state the intervals of concavity using interval notation. [5 Marks]

$$f'(x) = -3x^{-\frac{1}{3}} + 3x^{-\frac{2}{3}}$$

$$f''(x) = x^{-\frac{4}{3}} - 2x^{-\frac{5}{3}}$$

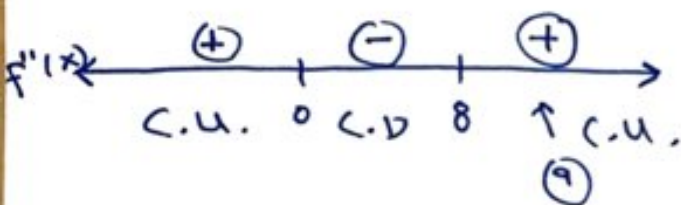
$$f''(x) = x^{-\frac{5}{3}}(x^{\frac{1}{3}} - 2)$$

$$\therefore \text{poi @ } (0, 0)$$

$$f''(x) = 0 \Rightarrow x = 8$$

$$f''(x) \text{ DNE} \Rightarrow x = 0$$

$$\begin{matrix} \downarrow \\ (8, 0) \end{matrix}$$



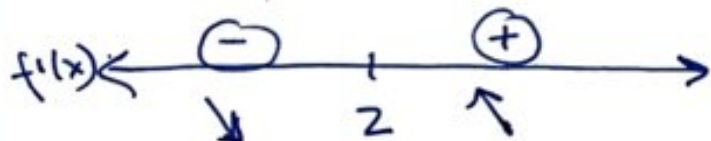
$$\text{c.u. } (-\infty, 0) \cup (8, \infty)$$

$$\text{c.d. } (0, 8)$$

8. Determine the coordinates of the absolute extrema of the function $f(x) = \sqrt{x^2 - 4x + 13}$ on the interval $[0, 6]$. [5 Marks]

$$f'(x) = \frac{1}{2}(x^2 - 4x + 13)^{-\frac{1}{2}}(2x - 4)$$

$$\text{critical \# @ } x = 2$$



$$\text{Local Min @ } (2, 3)$$

Test

$$f(2) = 3$$

$$f(0) = \sqrt{13}$$

$$f(6) = 5$$

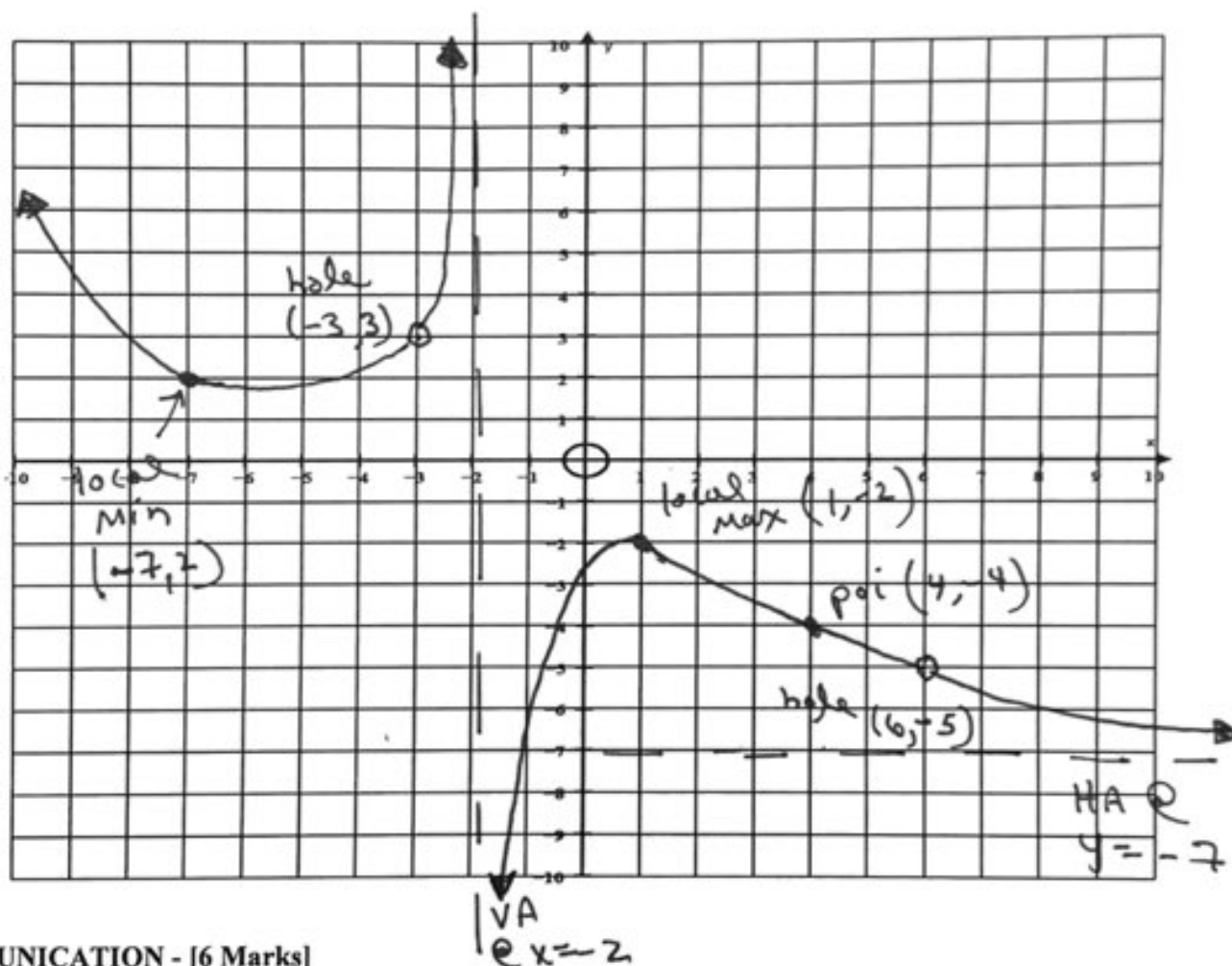
$$\therefore \text{Absolute Max @ } (6, 5)$$

$$\text{Absolute Min @ } (2, 3)$$

THINKING - [5 Marks]

1. Sketch the graph of **any** function having **all** of the following properties. **Note:** Label all holes, asymptotes, local extrema, cusps, vertical tangents and points of inflection. **[5 Marks]**

- $f(-7) = 2$, $f(1) = -2$, $f(4) = -4$, $f(-3)$ is undefined, $f(6)$ is undefined
- $f'(x) < 0$ for $x \in (-\infty, -7) \cup (1, 6) \cup (6, \infty)$, $f'(x) > 0$ for $x \in (-7, -3) \cup (-3, -2) \cup (-2, 1)$
- $f'(-7) = f'(1) = 0$
- $f''(x) > 0$ for $x \in (-\infty, -3) \cup (-3, -2) \cup (4, 6) \cup (6, \infty)$, $f''(x) < 0$ for $x \in (-2, 4)$
- $f''(4) = 0$
- $\lim_{x \rightarrow -2^-} f(x) = +\infty$, $\lim_{x \rightarrow -2^+} f(x) = -\infty$
- $\lim_{x \rightarrow -3} f(x) = 3$, $\lim_{x \rightarrow 6} f(x) = -5$
- $\lim_{x \rightarrow -\infty} f(x) = +\infty$, $\lim_{x \rightarrow +\infty} f(x) = -7$



COMMUNICATION - [6 Marks]

1a. A function, $f(x)$ has the property that $f'(c) = 0$. Does this guarantee that $f(x)$ has a local extremum at $x = c$? Briefly explain your answer. **[2 Marks]**

NO! There has to be a change from increasing to decreasing or vice versa.

1b. A function, $f(x)$ has the property that $f'(x) > 0$ on the interval (a, ∞) . Does this guarantee that $f(x)$ has a local minimum at $x = a$. Briefly explain your answer. **[2 Marks]**

NO! There has to be a change from increasing to decreasing. (i.e. $f'(x) < 0$ for $x < a$).

UNIT 3 ASSESSMENT OF LEARNING: CURVE SKETCHING - DAY 2

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App.	Thinking	Comm.
/20	/5	/2

APPLICATION - [20 Marks]

1. Sketch and properly label a graph of $f(x) = \frac{8(x^2-1)}{(x+3)^2}$. Note: $f'(x) = \frac{16(3x+1)}{(x+3)^3}$, $f''(x) = \frac{-96(x-1)}{(x+3)^4}$.

[12 Marks]

① Domain $(-\infty, -3) \cup (-3, \infty)$

② no holes

③ x-int @ $(\pm 1, 0)$; y-int $(0, -\frac{8}{9})$

④ VA @ $x = -3$

⑤ $\lim_{x \rightarrow -3^-} \frac{8(x^2-1)}{(x+3)^2} = \infty$; $\lim_{x \rightarrow -3^+} \frac{8(x^2-1)}{(x+3)^2} = \infty$

⑥ HA @ $y = 8$

⑧ crit. #'s @ $x = -\frac{1}{3}$

⑦ Cross test

$\frac{8(x^2-1)}{(x+3)^2} = 8$

$x^2-1 = x^2+6x+9$

$x = -\frac{5}{3}$

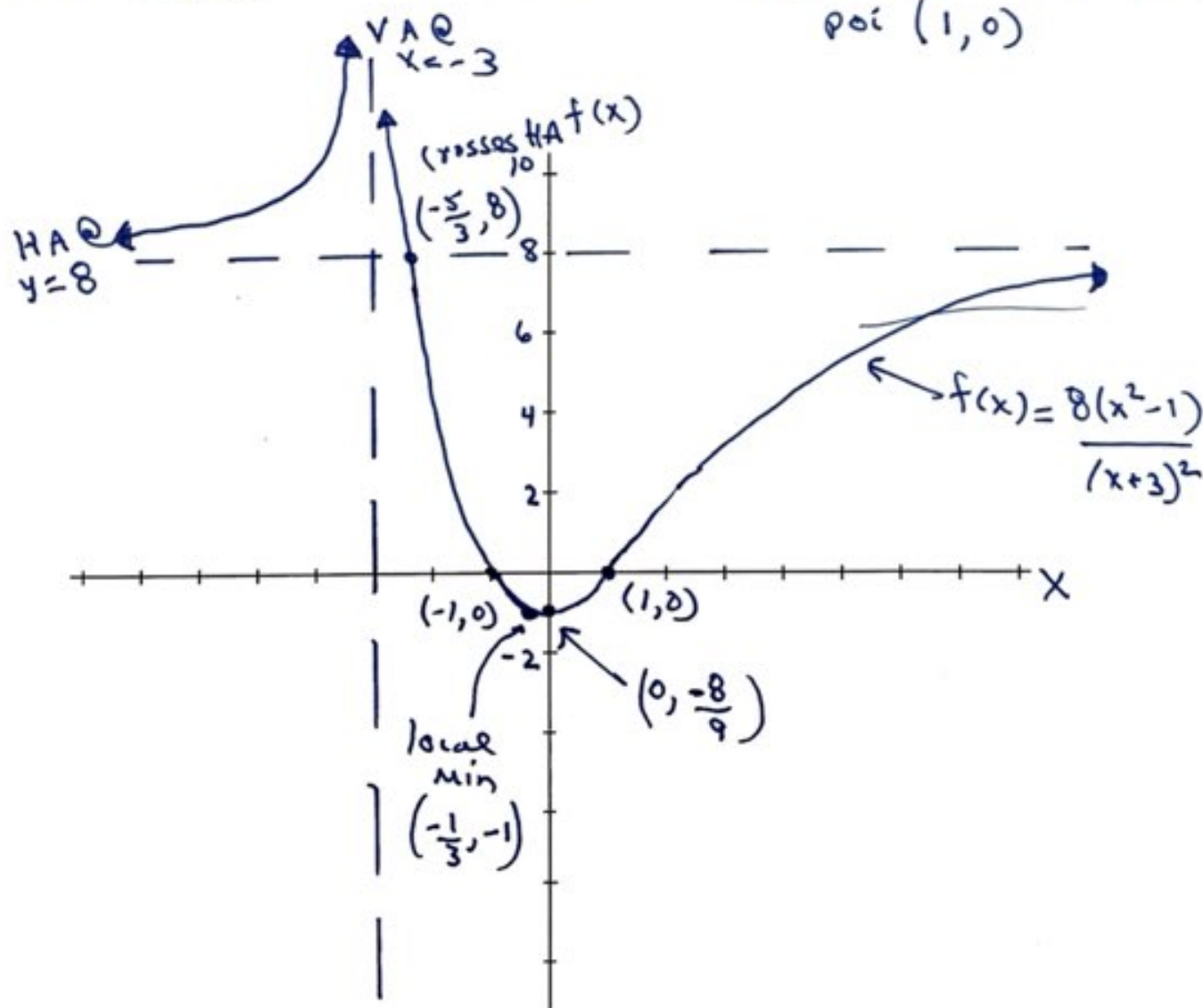
crosses @ $(-\frac{5}{3}, 8)$

⑨ local min @ $(-\frac{1}{3}, -1)$

⑩ poi & concavity
crit # @ $x = 1$

	$(-\infty, -3)$	$(-3, 1)$	$(1, \infty)$
$f''(x)$	+	+	-
$f(x)$	c.u	c.u	c.d

poi $(1, 0)$



2. Use the 2nd Derivative Test to determine the exact coordinates of the local extrema of the function $f(x) = x^3 + 2x^2 + x - 1$. Identify all the points as a local maximum or as a local minimum. [3 Marks]

$$f'(x) = 3x^2 + 4x + 1$$

$$= (3x+1)(x+1)$$

$$\text{crit \# } x = -\frac{1}{3}, -1$$

$$f''(x) = 6x + 4$$

$$f''(-\frac{1}{3}) = 2 > 0 \Rightarrow \text{local min at } (-\frac{1}{3}, -\frac{31}{27})$$

$$f''(-1) = -2 < 0 \Rightarrow \text{local max at } (-1, -1)$$

3. Determine the values of a and b if $f(x) = \frac{ax^2}{x-b}$, $a \& b > 0$ has a local minimum $(4, 32)$. [5 Marks]

$$f(4) = \frac{16a}{4-b}$$

$$f'(x) = \frac{2ax(x-b) - ax^2}{(x-b)^2}$$

$$32(4-b) = 16a$$

$$16a + 32b = 128$$

$$\boxed{a + 2b = 8} \quad (1)$$

$$f'(x) = \frac{2ax^2 - ax^2 - 2abx}{(x-b)^2}$$

$$f'(4) = 0$$

$$0 = 16a - 8ab$$

$$\boxed{2a - ab = 0} \quad (2)$$

$$\text{from (1)}$$

$$b = \frac{8-a}{2}$$

$$\text{sub } b \text{ into (2)}$$

$$2a - a\left(\frac{8-a}{2}\right) = 0$$

$$\frac{a^2}{2} - 2a = 0 \Rightarrow \begin{cases} a = 4 \\ b = 2 \end{cases}$$

THINKING - [5 Marks]

1. Determine the values of a, b, c, d and n so that the function $g(x) = \frac{ax^n + bx + c}{x+d}$ has a linear oblique asymptote of $y = 3x + 1$, a local maximum at $x = -3$ and a local minimum at $x = 11$. [5 Marks]

$$\begin{cases} n = 2 \\ a = 3 \end{cases}$$

$$g'(x) = \frac{(6x+b)(x+d) - (3x^2+bx+c)}{(x+d)^2}$$

$$= \frac{3x^2 + 6xd + bd - c}{(x+d)^2}$$

$$\begin{array}{r} * \quad \frac{3x+b-3d}{x+d} \overline{) 3x^2+bx+c} \\ \underline{-(3x^2+3dx)} \\ (b-3d)x+c \\ \underline{-(b-3d)x} \\ \Rightarrow \boxed{b-3d=1} \quad (1) \end{array}$$

$$* \quad f'(-3) = 0$$

$$27 - 18d + bd - c = 0 \quad (2)$$

$$\textcircled{3} - \textcircled{2}$$

$$324 + 84d = 0 \Rightarrow \boxed{d = -4}$$

$$\text{sub } d = -4 \text{ into (1)} \Rightarrow \boxed{b = -11}$$

$$\text{sub } d = -4, b = -11 \text{ into (2)}$$

$$27 + 72 + 44 - c = 0$$

$$\boxed{c = 143}$$

$$* \quad f'(11) = 0$$

$$363 + 66d + bd - c = 0 \quad (3)$$

$$\therefore \begin{cases} n = 2 \\ a = 3 \\ b = -11 \\ d = -4 \end{cases}$$

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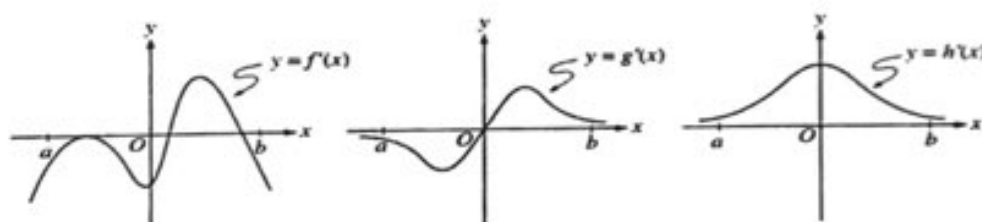
K & U	Thinking	Comm.
/16	/5	/6

KNOWLEDGE & UNDERSTANDING - [16 Marks]

Multiple Choice: Write the CAPITAL LETTER corresponding to the correct answer on the line provided.

[6 Marks]

- The function $f(x) = -\frac{1}{3}x^3 + x + 31$ is increasing on the interval(s) D
 A. $(-\infty, -1)$ B. $(-\infty, \infty)$ C. $(-\infty, -1) \cup (1, \infty)$ D. $(-1, 1)$
- Which of the following is/are true? A
 I. The function $f(x) = \frac{x-1}{x^4-16}$ has a vertical tangent.
 II. If $f'(c) = 0$ and $f''(c) < 0$ then $f(x)$ has a local maximum $x = c$.
 III. The function $f(x) = \frac{x^2-1}{x^2-4}$ has critical points at $x = \pm 2$.
 IV. If a function has a cusp at $x = -\frac{1}{7}$, then there is a point of inflection at $x = -\frac{1}{7}$.
 A. II only B. II, III and IV only C. I, and II only D. All are true.
- The critical number(s) of $f(x) = 4x^5 - 5x^3$ are A
 A. $0, \pm \frac{\sqrt{3}}{2}$ B. $\pm \frac{\sqrt{3}}{2}$ C. 0 D. There are none.
- The graph of the derivatives of $f(x)$, $g(x)$, and $h(x)$ are shown below. D



Which of the function(s) have a local minimum on $x \in (a, b)$?

- Which of the following is/are false regarding the graph below? C
 I. There is an absolute minimum at $(0, -4)$ and local maximums at $(-2, 0)$ and $(2, 0)$.
 II. There is a local minimum at $(0, -4)$ and absolute minimums at $(-3, -5)$ and $(3, -5)$.
 III. There are local minimums at $(0, -4)$, $(-3, -5)$ and $(3, -5)$.
 IV. Points $(-2, 0)$ and $(2, 0)$ are both local and absolute maximums.
 A. II and IV only B. II and III only C. I and III only D. I, II and III only
- The value of the constant b for the function $f(x) = x^2 + bx + 4$, which has a critical point at $x = -4$ is D
 A. 2 B. 4 C. 6 D. 8

7. For the function $f(x) = \frac{2x^2}{x^2+1}$, determine the **exact** coordinates of the points of inflection and state the intervals of concavity using interval notation. [5 Marks]

$$f'(x) = \frac{4x(x^2+1) - 2x^2(2x)}{(x^2+1)^2} \quad f''(x) = \frac{4(x^2+1)^2 - (4x)(2(x^2+1)(2x))}{(x^2+1)^4}$$

$$f'(x) = \frac{4x^3 + 4x - 4x^3}{(x^2+1)^2}$$

$$= \frac{4(x^2+1)[(x^2+1) - 4x^2]}{(x^2+1)^4}$$

$$f'(x) = \frac{4x}{(x^2+1)^2}$$

$$= \frac{-12x^2 + 4}{(x^2+1)^3}$$

$$= \frac{-4(3x^2 - 1)}{(x^2+1)^3}$$

$f''(x) \leftarrow \begin{array}{c} (-) \quad (+) \quad (-) \\ \hline \frac{-1}{\sqrt{3}} \quad \frac{1}{\sqrt{3}} \end{array} \rightarrow$
 $\therefore \text{p.o.i. @ } \left(\frac{-1}{\sqrt{3}}, \frac{1}{2}\right) \text{ \& } \left(\frac{1}{\sqrt{3}}, \frac{1}{2}\right)$
 C.D. $\left(\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$
 C.D. $\left(-\infty, \frac{-1}{\sqrt{3}}\right) \cup \left(\frac{1}{\sqrt{3}}, \infty\right)$

8. Determine the coordinates of the absolute extrema of the function $f(x) = x^3 - 6x^2 + 4$ on the interval $[-2, 5]$. [5 Marks]

$$f'(x) = 3x^2 - 12x$$

$$= 3x(x-4)$$

critical # @ $x=0, x=4$

$f''(x) \leftarrow \begin{array}{c} (+) \quad (-) \quad (+) \\ \hline \uparrow \quad 0 \quad \downarrow \quad 4 \quad \uparrow \end{array} \rightarrow$

Local min @ $(4, -28)$

Local max @ $(0, 4)$

$$f(4) = -28$$

$$f(0) = 4$$

$$f(-2) = -28$$

$$f(5) = -21$$

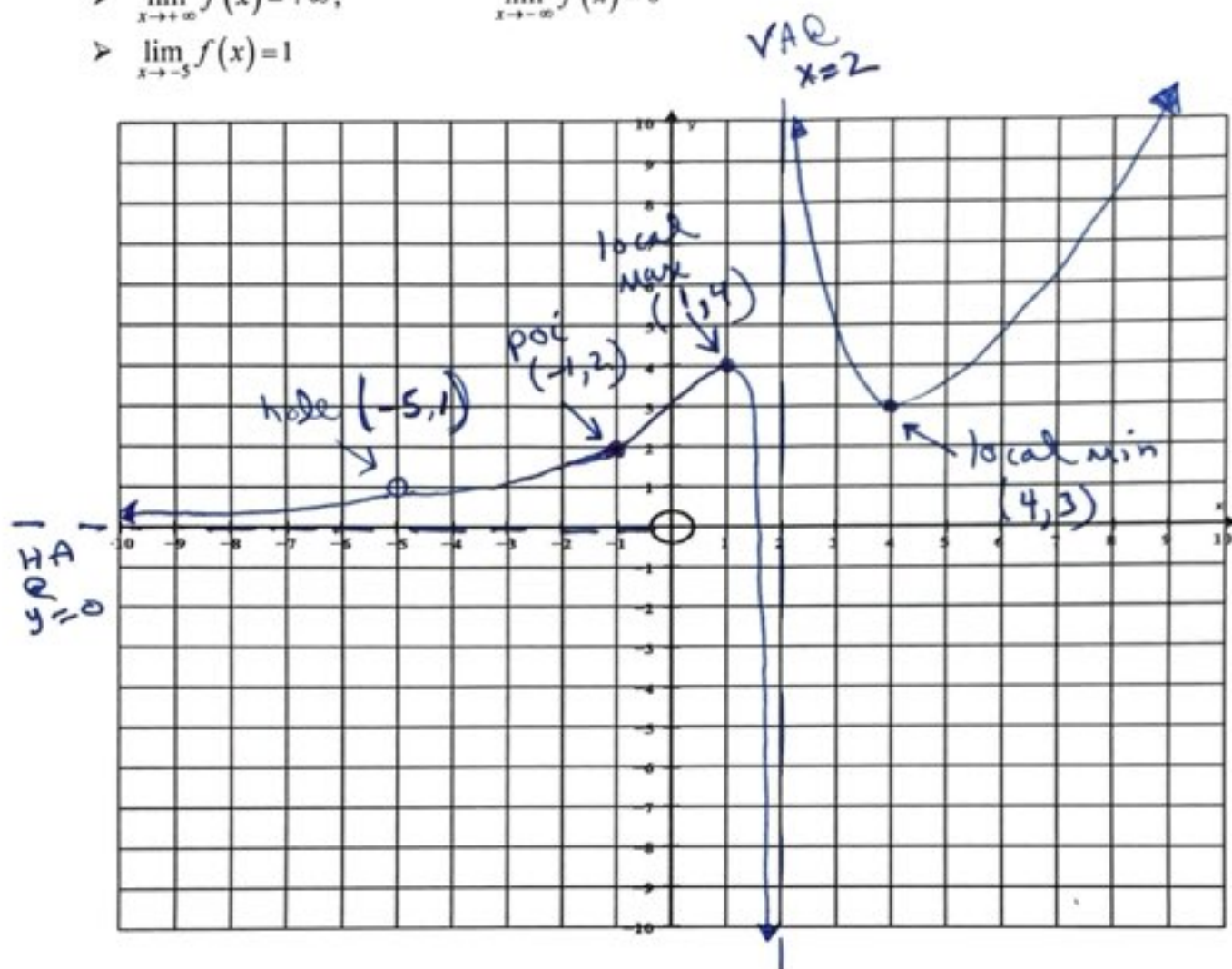
Absolute min @ $(4, -28) \& (-2, -28)$

Absolute max @ $(0, 4)$

THINKING - [5 Marks]

1. Sketch the graph of **any** function having **all** of the following properties. **Note:** Label all holes, asymptotes, local extrema, cusps, vertical tangents and points of inflection. [5 Marks]

- $f(-1) = 2$, $f(1) = 4$, $f(4) = 3$, $f(-5)$ is undefined
- $f'(x) > 0$ for $x \in (-\infty, -5) \cup (-5, 1) \cup (4, \infty)$, $f'(x) < 0$ for $x \in (1, 2) \cup (2, 4)$, $f'(1) = f'(4) = 0$
- $f''(x) < 0$ for $x \in (-1, 2)$, $f''(x) > 0$ for $x \in (-\infty, -5) \cup (-5, -1) \cup (2, \infty)$, $f''(-1) = 0$
- $\lim_{x \rightarrow 2^-} f(x) = -\infty$, $\lim_{x \rightarrow 2^+} f(x) = +\infty$
- $\lim_{x \rightarrow +\infty} f(x) = +\infty$, $\lim_{x \rightarrow -\infty} f(x) = 0$
- $\lim_{x \rightarrow -5} f(x) = 1$



COMMUNICATION - [6 Marks]

- 1a. A function, $f(x)$ has the property that $f''(c) = 0$. Does this guarantee that $f(x)$ has a point of inflection at $x = c$? Briefly explain your answer. [2 Marks]

No! There has to be a change in concavity at $x = c$.

- 1b. A function, $f(x)$ has the property that $f''(x) > 0$ on the interval $(-\infty, a)$ and $f''(x) < 0$ on the interval (a, ∞) . Does this guarantee that $f(x)$ has a point of inflection at $x = a$. Briefly explain your answer. [2 Marks]

No! There may be a vertical asymptote or a hole at $x = a$. (ie: $f(a)$ must be defined)

UNIT 3 ASSESSMENT OF LEARNING: CURVE SKETCHING - DAY 2

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App.	Thinking	Comm.
/20	/5	/2

APPLICATION - [20 Marks]

1. Sketch and properly label a graph of $f(x) = \frac{x^3}{x^2-4}$. Note: $f'(x) = \frac{x^2(x^2-12)}{(x^2-4)^2}$ and $f''(x) = \frac{8x(x^2+12)}{(x^2-4)^3}$.

[12 Marks]

① Domain $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$

② no holes

③ x-int @ $(0,0)$; y-int $(0,0)$

④ VA @ $x = \pm 2$

⑤ $\lim_{x \rightarrow -2^-} f(x) = -\infty$ $\lim_{x \rightarrow 2^-} f(x) = -\infty$
 $\lim_{x \rightarrow -2^+} f(x) = \infty$ $\lim_{x \rightarrow 2^+} f(x) = \infty$

⑥ $\frac{x^3}{x^2-4} = \frac{x}{4x} + \frac{x}{x^2-4}$
 $OA @ y=x$

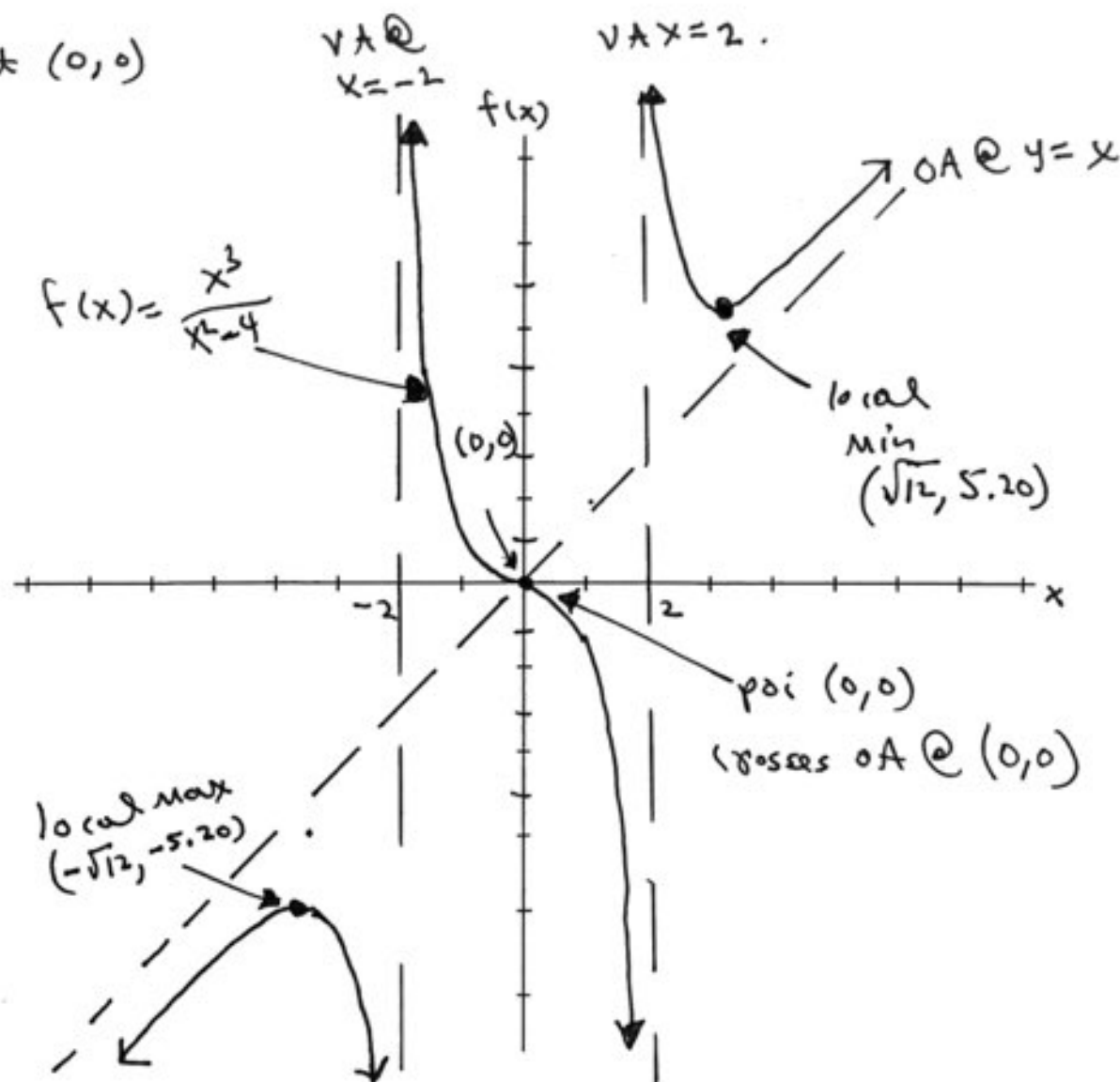
⑦ crit #: $0, \pm\sqrt{12}$

$(-\infty, -\sqrt{12})$	$(-\sqrt{12}, -2)$	$(-2, 0)$	$(0, 2)$	$(2, \sqrt{12})$	$(\sqrt{12}, \infty)$
$f'(x)$	+	-	-	-	+
$f(x)$	↗	↘	↘	↘	↗

⑧ poi & concavity
crit #: $x=0$

$(-\infty, -2)$	$(-2, 0)$	$(0, 2)$	$(2, \infty)$
$f''(x)$	-	+	-
$f(x)$	c.d.	c.u.	c.d.

⑨ cross test
 $4x=0$
 $x=0$
crosses at $(0,0)$



2. Determine the values of a and b if $f(x) = \frac{ax^2+b}{x^3}$ has a local maximum at $(\sqrt{3}, \frac{14\sqrt{3}}{9})$. [5 Marks]

$$\begin{aligned} f'(x) &= \frac{2ax(x^3) - (ax^2+b)(3x^2)}{x^6} \\ &= \frac{x^2(2ax^2 - 3ax^2 - 3b)}{x^6} \\ &= \frac{-ax^2 - 3b}{x^4} \end{aligned}$$

$$f(\sqrt{3}) = \frac{14\sqrt{3}}{9}$$

$$\frac{3a+b}{3\sqrt{3}} = \frac{14\sqrt{3}}{9}$$

$$27a + 9b = 126$$

$$\boxed{3a + b = 14} \quad (2)$$

$$f'(\sqrt{3}) = 0$$

$$\frac{-3a - 3b}{9} = 0$$

$$\boxed{a + b = 0} \quad (1)$$

$$(2) - (1)$$

$$2a = 14$$

$$a = 7$$

sub $a = 7$ into (1)

$$b = -7$$

3. Use the 2nd Derivative Test to determine the exact coordinates of the local extrema of the function

$f(x) = \frac{2}{3}x^3 + 4x^2 - 24x - 10$. Identify all the points as a local maximum or as a local minimum. [3 Marks]

$$f'(x) = 2x^2 + 8x - 24$$

$$f''(x) = 4x + 8$$

$$= 2(x^2 + 4x - 12)$$

$$= 2(x+6)(x-2)$$

$$\text{crit \# } x = -6, x = 2.$$

$$f''(-6) = -16 < 0 \Rightarrow \text{local max}$$

$$(-6, 134)$$

$$f''(2) = 16 > 0 \Rightarrow \text{local min}$$

$$(2, -\frac{110}{3})$$

THINKING - [5 Marks]

1. Given $f(x) = 4x^4 - 24x^2 - 1$, determine the point of intersection of the tangents to the curve at the function's points of inflection. Note: You must verify the points of inflection. [5 Marks]

$$f'(x) = 16x^3 - 48x$$

$$f''(x) = 48x^2 - 48$$

$$\begin{array}{ccc} (+) & (-) & (+) \\ \leftarrow & & \rightarrow \\ \text{C.U.} & \text{C.D.} & \text{C.U.} \end{array}$$

$$= 48(x-1)(x+1) \Rightarrow x = \pm 1 \quad \therefore \text{poi @ } (1, -21) \text{ \& } (-1, -21)$$

$$f'(1) = -32$$

$$f'(-1) = 32$$

$$\Rightarrow y + 21 = -32(x-1) \quad (1)$$

$$y + 21 = 32(x+1) \quad (2)$$

$$(1) - (2)$$

$$0 = -64x$$

$$x = 0$$

sub $x = 0$ into (1)

$$y + 21 = 32$$

$$y = 11$$

$$\therefore \text{point of intersection } (0, 11)$$